

Parametric Models

All models are wrong, but some are useful.

—GEORGE BOX

We have thus far focused on making credible inferences based on an agnostic view of the world. That is, our inferences have not depended on having full (or nearly full) knowledge of how our data were generated. This approach has not been the historical norm in statistics or econometrics. Instead, these disciplines have primarily been developed around the idea of constructing statistical models to explain observed phenomena, and then imposing distributional assumptions to justify inferences within the framework of the model.

In this chapter, we show how these stronger distributional assumptions can be used profitably to assist in statistical inference, even when these assumptions do not exactly hold. Much as regression provides a method for estimating a best linear approximation of a potentially infinite-dimensional conditional expectation function, we will see that maximum likelihood estimation provides a method for estimating a “best” *parametric* approximation of a potentially infinite-dimensional cumulative distribution function. We begin by defining parametric models, and discussing some canonical examples.

5.1 MODELS AND PARAMETERS

A *parametric model* is a model that assumes that the distribution of some outcome variable Y , conditional on a set of explanatory variables $\mathbf{X}$, can be fully characterized by a finite number of *parameters*. We formalize this as follows:

Definition 5.1.1. Parametric Model

For a random variable Y and random vector $\mathbf{X}$ of length K , a *parametric model* is a set of functions $g : \mathbb{R}^{K+1} \rightarrow \mathbb{R}$ indexed by a real-valued

parameter vector θ of length τ :

$$\{g(y, \mathbf{x}; \theta) : \theta \in \Theta\},^1$$

where $\Theta \subseteq \mathbb{R}^\tau$. The set Θ is called the *parameter space*. The model is said to be *true* if $\exists \theta \in \Theta$ such that

$$f_{Y|X}(y|\mathbf{x}) = g(y, \mathbf{x}; \theta).$$

A model is simply a set of distributions. We label each distribution distinctly with θ , which is the *parameter vector* that describes that distribution. For a parametric model, θ is assumed to have a finite number of real-valued elements. This parametric approach contrasts with our approach in earlier chapters: for all our key results, we did not require restrictive distributional assumptions on any of the variables— g was allowed to be unknown and perhaps characterized by an infinite-dimensional θ .

To illustrate how parametric models work, we will start with an extremely simple example: the (potentially) biased coin that we considered in Example 1.2.7 (*A Biased Coin Flip*).

Example 5.1.2. A Biased Coin Flip

Let $Y = 0$ if the coin comes up heads and $Y = 1$ if the coin comes up tails. The only unknown quantity is the probability p that the coin turns up tails. Thus, Y is a Bernoulli random variable with probability p . Since there are no explanatory variables, we need only consider the (unconditional) PMF $f_Y(y)$. Let

$$f_Y(y) = g(y; p) = \begin{cases} 1 - p & : y = 0 \\ p & : y = 1 \\ 0 & : \text{otherwise.} \end{cases}$$

Then $\theta = p$, since p completely determines the distribution of the random variable Y . The parameter space is $\Theta = [0, 1]$, which is the unit interval. This is, technically, a parametric model for Y . If we knew just one number, p , then we would know everything there is to know about the distribution of Y . $\triangle$

In this setting, assuming a parametric model is unobjectionable, as the problem itself naturally only admits so much complexity. Similarly, consider a roll of a (potentially) loaded six-sided die.

¹ Two notes on notation: (1) We separate θ from the arguments of g with a semicolon. (2) When we have no explanatory variables, we will use as shorthand $f_Y(y) = g(y; \theta)$.

Example 5.1.3. A Loaded Die Roll

Let Y take on the value of the side of the die that comes up. Our model is

$$f_Y(y) = g(y; \mathbf{p}) = \begin{cases} p_1 & : y = 1 \\ p_2 & : y = 2 \\ p_3 & : y = 3 \\ p_4 & : y = 4 \\ p_5 & : y = 5 \\ p_6 & : y = 6 \\ 0 & : \text{otherwise,} \end{cases}$$

where $\mathbf{p} = (p_1, p_2, p_3, p_4, p_5, p_6)$. Then $\boldsymbol{\theta} = \mathbf{p}$, since those six values completely determine the distribution of Y . The parameter space is

$$\Theta = \left\{ \mathbf{p} \in \mathbb{R}^6 : \forall i, 0 \leq p_i \leq 1, \text{ and } \sum_{i=1}^6 p_i = 1 \right\},$$

that is, the components of $\mathbf{p}$ must each be between zero and one, and must all sum to one. This is, again, a parametric model for Y . Formally, Y follows a *categorical distribution* (also known as a *generalized Bernoulli distribution*) with event probabilities p_1, p_2, p_3, p_4, p_5 , and p_6 . Δ

Examples 5.1.2 and 5.1.3 represent the simplest types of statistical models. These models describe the (unconditional) distribution of a random variable Y in terms of a finite number of parameters. However, as Definition 5.1.1 suggests, some of the most important and widely used models characterize the *conditional* distribution of an outcome variable Y given a set of explanatory variables $\mathbf{X}$. We can now consider some of these more complicated models.

5.1.1 The Classical Linear Model

The *classical linear model* is the workhorse of conventional applied econometrics and statistics, and is typically afforded a great deal of attention in conventional textbooks. This model can be defined in accordance with our general definition of a parametric model as follows:

Definition 5.1.4. Classical Linear Model

For a random variable Y and random vector $\mathbf{X} = (1, X_{[1]}, X_{[2]}, \dots, X_{[K]})$, the *classical linear model* is a parametric model with

$$g(y, \mathbf{x}; (\boldsymbol{\beta}, \sigma)) = \phi(y; (\mathbf{x}\boldsymbol{\beta}, \sigma^2)),$$

where $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_K)^T$, $\phi(y; (\mu, \sigma^2))$ denotes the PDF of the normal distribution with mean μ and variance σ^2 , and the parameter space is $\Theta = \{(\boldsymbol{\beta}, \sigma) \in \mathbb{R}^{K+2} : \sigma \geq 0\}$.

Equivalently, the classical linear model can also be written in the following (much more common) form:

$$Y = \mathbf{X}\boldsymbol{\beta} + \varepsilon,$$

where $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_K)^T$ and $\varepsilon \sim N(0, \sigma^2)$. This representation illustrates the two key elements of the classical linear model: the response is linear in the explanatory variables, and errors are i.i.d. normal with expectation zero. The fact that the deviations from the linear component are mean zero and independent of $\mathbf{X}$ should be somewhat familiar; these are the properties of the deviations from the CEF (Theorem 2.2.19). The following theorem codifies this connection between the classical linear model and the CEF.

Theorem 5.1.5. *CEF of the Classical Linear Model*

For a random variable Y and random vector $\mathbf{X} = (1, X_{[1]}, X_{[2]}, \dots, X_{[K]})$, if the classical linear model is true, then the CEF of Y given $\mathbf{X}$ is

$$E[Y|\mathbf{X} = \mathbf{x}] = \mathbf{x}\boldsymbol{\beta}, \forall \mathbf{x} \in \text{Supp}[\mathbf{X}].$$

Proof: If the classical linear model is true, then $f_{Y|\mathbf{X}} = \phi(y; (\mathbf{x}\boldsymbol{\beta}, \sigma^2))$. So, by Definition 2.2.10 (*Conditional Expectation*), $\forall \mathbf{x} \in \text{Supp}[\mathbf{X}]$,

$$E[Y|\mathbf{X} = \mathbf{x}] = \int_{-\infty}^{\infty} y\phi(y; (\mathbf{x}\boldsymbol{\beta}, \sigma^2))dy = \mathbf{x}\boldsymbol{\beta},$$

since the second expression is just the expectation of a random variable with distribution $N(\mathbf{x}\boldsymbol{\beta}, \sigma^2)$. $\square$

Note that, as in Section 2.2.4, since the CEF is linear, the BLP is also the CEF here. Let us consider an example of the classical linear model in practice.

Example 5.1.6. *Modeling SAT Scores*

Suppose that we are trying to model SAT scores for high school students. We have the following information about each student i :

- Income_i : student i 's family income in the previous calendar year,
- Tutoring_i : an indicator for whether or not student i received tutoring,
- GPA_i : student i 's high school GPA, and
- Y_i : student i 's SAT score.

Let $\mathbf{X}_i = (1, \text{Income}_i, \text{Tutoring}_i, \text{GPA}_i)$. Then we might impose the following linear model:

$$Y_i = \mathbf{X}_i \boldsymbol{\beta} + \varepsilon_i \\ = \beta_0 + \beta_1 \cdot \text{Income}_i + \beta_2 \cdot \text{Tutoring}_i + \beta_3 \cdot \text{GPA}_i + \varepsilon_i,$$

where $\varepsilon_i \sim N(0, \sigma^2)$. The final term ε_i represents the “disturbance,” that is, all variation in SAT scores that is not explained by income, tutoring, GPA, and the constant.

In this example, the linear model implies that, if we knew just five numbers, we would know *everything* about the distribution of SAT scores conditional on income, tutoring, and GPA. It is a parametric model with parameter vector $\boldsymbol{\theta} = (\beta_0, \beta_1, \beta_2, \beta_3, \sigma)$. The conditional PDF of Y_i given $\mathbf{X}_i$ is assumed to be of the form

$$f_{Y_i|\mathbf{X}_i}(y_i|\mathbf{x}_i) = \phi(y_i; (\beta_0 + \beta_1 \cdot \text{Income}_i + \beta_2 \cdot \text{Tutoring}_i + \beta_3 \cdot \text{GPA}_i, \sigma^2)). \triangle$$

Defenders of the classical linear model would argue that “all the things that we do not measure probably add up to a normal distribution” and that the assumption that $\varepsilon_i \sim N(0, \sigma^2)$ is a harmless technical convenience. It is true that assuming a normally distributed disturbance is not entirely unreasonable, since the Central Limit Theorem implies that the sum of many small, independent, random perturbations will tend to be normally distributed. But, at the same time, we would be hard-pressed to accept this assumption literally, much as we would be unlikely to accept an assumption of linearity literally in most settings.

5.1.2 Binary Choice Models

Let us consider another class of commonly used statistical models: *binary choice models*. Binary choice models allow for the prediction of outcomes that are governed by a Bernoulli distribution (Example 1.2.7). In short, a binary choice model assumes that the conditional probability associated with the Bernoulli distribution is governed by a known function of the explanatory variables. We provide a formal definition below.

Definition 5.1.7. Binary Choice Model²

For a binary random variable Y and random vector $\mathbf{X} = (1, X_{[1]}, X_{[2]}, \dots, X_{[K]})$, a *binary choice model* is a parametric model with

$$g(y, \mathbf{X}; \boldsymbol{\beta}) = \begin{cases} 1 - h(\mathbf{X}\boldsymbol{\beta}) & : y = 0 \\ h(\mathbf{X}\boldsymbol{\beta}) & : y = 1 \\ 0 & : \text{otherwise,} \end{cases}$$

where $\beta = (\beta_0, \beta_1, \dots, \beta_K)^T$, the mean function $h: \mathbb{R} \rightarrow [0, 1]$ is known, and the parameter space is $\Theta = \mathbb{R}^{K+1}$.

The function h is called the *mean function*³ because it governs the shape of the CEF. This is codified in the following theorem.

Theorem 5.1.8. *CEF of a Binary Choice Model*

For a binary random variable Y and random vector $\mathbf{X} = (1, X_{[1]}, X_{[2]}, \dots, X_{[K]})$, if the binary choice model with mean function h is true, the CEF of Y given $\mathbf{X}$ is

$$E[Y|\mathbf{X} = \mathbf{x}] = h(\mathbf{X}\beta), \forall \mathbf{x} \in \text{Supp}[\mathbf{X}].$$

A proof follows from Example 2.1.3. Importantly, one consequence of a binary choice model is that all predictions of the CEF must be bounded in $[0, 1]$, since the mean function has codomain $[0, 1]$. Note that, importantly, the CEF may be nonlinear in $\mathbf{X}$. Linear approximations of the CEF (such as the BLP) need not respect the bounds of the interval $[0, 1]$; predictions of conditional expectations from such approaches might yield negative predictions or predictions greater than one. Such predictions would be nonsensical, and in fact are incompatible with the Kolmogorov axioms (Definition 1.1.2). In settings where respecting the bounds of predictions is important (we will have such a case in Section 7.2.3), we will generally prefer to use a binary choice model rather than an approach (like regression) that might yield nonsensical predictions.

The two most commonly used binary choice models are *logit* and *probit*, which differ only in the shape of their mean function. We define both models below.

Definition 5.1.9. *Logit Model*

For a binary random variable Y and random vector $\mathbf{X} = (1, X_{[1]}, X_{[2]}, \dots, X_{[K]})$, a *logit model* is a binary choice model where the mean function is the logistic function:

$$h(\mathbf{X}\beta) = \frac{e^{\mathbf{X}\beta}}{1 + e^{\mathbf{X}\beta}}.$$

² Technically, we are restricting our attention to a special type of binary choice model: *index models*. Index models assume that the explanatory variables enter into the parametric model only through the linear *index* $\mathbf{X}\beta$. In practice, this can be relaxed in many ways, including the polynomials and interactions that we considered in Section 4.3 (*Estimation of Nonlinear Conditional Expectation Functions*).

³ The *link function*, or the inverse of the mean function, is conventionally how such models are parameterized.

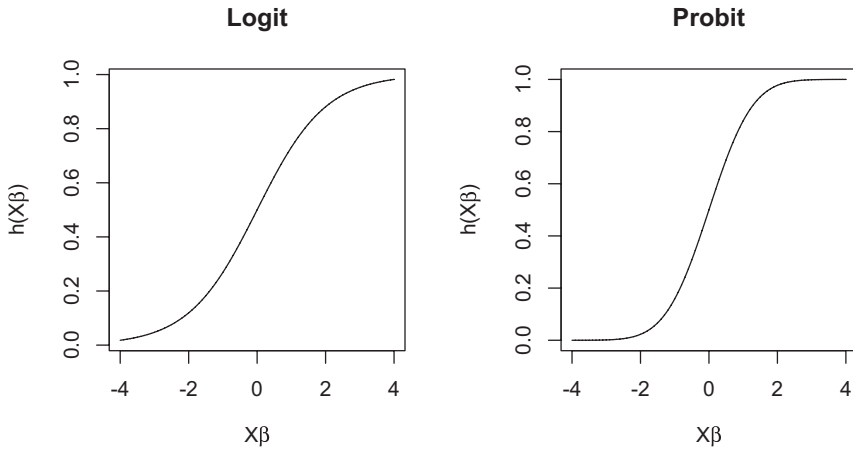


FIGURE 5.1.1. *Logit and Probit Functions*

Definition 5.1.10. *Probit Model*

For a binary random variable Y and random vector $\mathbf{X} = (1, X_{[1]}, X_{[2]}, \dots, X_{[K]})$, a *probit model* is a binary choice model where the mean function is the CDF of the standard normal distribution:

$$h(\mathbf{X}\boldsymbol{\beta}) = \Phi(\mathbf{X}\boldsymbol{\beta}).$$

Figure 5.1.1 displays the graphs of these functions.

Note that the mean functions for logit and probit have very similar shapes, differing primarily in scale.⁴ In both models, large changes in $\mathbf{X}\boldsymbol{\beta}$ at extreme values correspond to relatively small changes in the conditional probability of $Y=1$.

We conclude this section with an example of how binary choice models work in practice.

Example 5.1.11. *Modeling the Decision to Buy a Car*

Suppose that we are trying to model the probability that an individual will buy a car. We have the following information about each individual i :

- *Income_{*i*}*: individual i 's income in the previous calendar year.
- *Distance_{*i*}*: individual i 's distance from their workplace.
- *PublicTransit_{*i*}*: an indicator for whether or not individual i has access to public transportation.
- *Y_{*i*}*: an indicator for whether or not individual i bought a car.

⁴ As a result, for most reasonable data generating processes, conclusions drawn using logit and probit models should not differ much.

Let $\mathbf{X}_i = (1, \text{Income}_i, \text{Distance}_i, \text{PublicTransit}_i)$. Then we might assume that

$$\begin{aligned}\Pr[Y_i = 1 | \mathbf{X}_i] &= h(\mathbf{X}_i \boldsymbol{\beta}) \\ &= h(\beta_0 + \beta_1 \cdot \text{Income}_i + \beta_2 \cdot \text{Distance}_i + \beta_3 \cdot \text{PublicTransit}_i).\end{aligned}$$

Once we assume the form of h , this is a parametric model: assuming the model is true, knowledge of $\boldsymbol{\beta}$ tells us everything about the conditional distribution of Y_i given $\mathbf{X}_i$. For example, if h is the standard normal CDF, then we have a probit model with

$$\begin{aligned}\Pr[Y_i = 1 | \mathbf{X}_i] &= \Phi(\mathbf{X}_i \boldsymbol{\beta}) \\ &= \Phi(\beta_0 + \beta_1 \cdot \text{Income}_i + \beta_2 \cdot \text{Distance}_i + \beta_3 \cdot \text{PublicTransit}_i).\end{aligned}$$

If β_0 , β_1 , β_2 , and β_3 are known, this equation fully describes the conditional distribution of Y_i given $\mathbf{X}_i$. Δ

5.2 MAXIMUM LIKELIHOOD ESTIMATION

Once we have chosen a parametric model to describe an observed phenomenon, how can we estimate its parameters from the data? As usual, we will assume that we have independent and identically distributed (i.i.d.) observations. It can then be shown that, if we assume that the parametric model is true, then a procedure known as *maximum likelihood (ML) estimation* yields consistent, asymptotically efficient, and asymptotically normal estimates of the parameters. Indeed, under this assumption, there exists no “regular” estimator that has better asymptotic efficiency. However, even when the parametric model is not true, we will show that ML estimation nevertheless yields an approximation that is sensible and can be used profitably for plug-in estimation.

5.2.1 The Logic of Maximum Likelihood Estimation

Suppose that we have n i.i.d. observations of a discrete random vector $(Y, \mathbf{X})$, and suppose that we assume the model

$$f_{Y|\mathbf{X}}(y|\mathbf{x}) = g(y, \mathbf{x}; \boldsymbol{\theta}),$$

so that the conditional PMF of Y given $\mathbf{X}$ is entirely characterized by some known function g and the parameter vector $\boldsymbol{\theta}$. The idea behind ML estimation is that we want to find the parameters $\boldsymbol{\theta}$ that would have maximized the probability of obtaining the data that we actually observed. In other words, since we do not know the parameters but we *do* know the outcome—or rather, n i.i.d. outcomes—our best guess for the parameters is

the values that would have made that particular set of outcomes as likely as possible.

To formally describe ML estimation, we must first define the *likelihood function*.

Definition 5.2.1. Likelihood Function

Let Y be a random variable and $\mathbf{X}$ be a random vector, and assume the parametric model $f_{Y|\mathbf{X}}(y|\mathbf{x}) = g(y, \mathbf{x}; \boldsymbol{\theta})$. Then the *likelihood function* is

$$\mathcal{L}(\mathbf{t} | Y, \mathbf{X}) = g(Y, \mathbf{X}; \mathbf{t}).$$

Thus, if the model is true, the likelihood function maps each candidate parameter value $\mathbf{t}$ to the conditional probability/density of the observed values of Y given the observed values of $\mathbf{X}$, assuming $\mathbf{t}$ is the true parameter value.⁵

We can generalize this to the case where we have i.i.d. observations. Let $(Y_1, \mathbf{X}_1), (Y_2, \mathbf{X}_2), \dots, (Y_n, \mathbf{X}_n)$ be i.i.d. observations of $(Y, \mathbf{X})$, and let

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} \quad \text{and} \quad \mathbb{X} = \begin{pmatrix} X_{[1]1} & X_{[2]1} & \cdots & X_{[K]1} \\ X_{[1]2} & X_{[2]2} & \cdots & X_{[K]2} \\ \vdots & \vdots & \ddots & \vdots \\ X_{[1]n} & X_{[2]n} & \cdots & X_{[K]n} \end{pmatrix}.$$

Then

$$\mathcal{L}(\mathbf{t} | \mathbf{Y}, \mathbb{X}) = \prod_{i=1}^n \mathcal{L}(\mathbf{t} | Y_i, \mathbf{X}_i).$$

We can now formally define the maximum likelihood estimator.

Definition 5.2.2. Maximum Likelihood Estimator

Let Y be a random variable and $\mathbf{X}$ be a random vector. Let $(Y_1, \mathbf{X}_1), (Y_2, \mathbf{X}_2), \dots, (Y_n, \mathbf{X}_n)$ be i.i.d. observations of $(Y, \mathbf{X})$, and let

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} \quad \text{and} \quad \mathbb{X} = \begin{pmatrix} X_{[1]1} & X_{[2]1} & \cdots & X_{[K]1} \\ X_{[1]2} & X_{[2]2} & \cdots & X_{[K]2} \\ \vdots & \vdots & \ddots & \vdots \\ X_{[1]n} & X_{[2]n} & \cdots & X_{[K]n} \end{pmatrix}.$$

⁵ Formally, we would say that the likelihood function is itself a random function, $\mathcal{L} : \Omega \rightarrow \mathcal{L}$, where $\mathcal{L}$ is the space of functions $L : \mathbb{R} \rightarrow [0, 1]$.

Then, for the parametric model $f_{Y|X}(y|\mathbf{x}) = g(y, \mathbf{x}; \boldsymbol{\theta})$ with $\boldsymbol{\theta} \in \boldsymbol{\Theta}$, the *maximum likelihood estimator* of $\boldsymbol{\theta}$ is

$$\hat{\boldsymbol{\theta}}_{ML} = \arg \max_{\mathbf{t} \in \boldsymbol{\Theta}} \mathcal{L}(\mathbf{t} | \mathbf{Y}, \mathbb{X}) = \arg \max_{\mathbf{t} \in \boldsymbol{\Theta}} \prod_{i=1}^n \mathcal{L}(\mathbf{t} | Y_i, \mathbf{X}_i),$$

or, equivalently,

$$\hat{\boldsymbol{\theta}}_{ML} = \arg \max_{\mathbf{t} \in \boldsymbol{\Theta}} [\log \mathcal{L}(\mathbf{t} | \mathbf{Y}, \mathbb{X})] = \arg \max_{\mathbf{t} \in \boldsymbol{\Theta}} \sum_{i=1}^n \log \mathcal{L}(\mathbf{t} | Y_i, \mathbf{X}_i),^6$$

where $\log \mathcal{L}(\mathbf{t} | \mathbf{Y}, \mathbb{X})$ is known as the *log-likelihood*.

The equivalence of the two formulations of the ML estimator arises because the natural logarithm—which we write as $\log$ —is a strictly increasing function; therefore, the maximum log-likelihood must be attained at the same point as the maximum likelihood. Depending on the type of problem we are considering, we might prefer one formulation over another, though typically the log-likelihood is preferred as sums tend to be more analytically tractable than products.

Readers may note a certain similarity of the ML estimator to the form of the OLS regression estimator (Definition 4.1.2): both estimators can be written as optimization problems. The OLS estimator minimizes the sum of squared residuals, and the ML solution maximizes the sum over every unit's log-likelihood function.⁷

To ground these ideas, we will consider two simple examples. We begin with our usual coin flip example.

Example 5.2.3. Maximum Likelihood Estimation for a Biased Coin Flip

As usual, let $Y = 0$ if the coin comes up heads and $Y = 1$ if the coin comes up tails. Our model is

$$f_Y(y) = g(y; p) = \begin{cases} 1 - p & : y = 0 \\ p & : y = 1 \\ 0 & : \text{otherwise,} \end{cases}$$

with $\boldsymbol{\Theta} = [0, 1]$. The one thing we do not know is the parameter p , the probability that the coin turns up tails. Suppose that we observed n i.i.d. coin flips: $\mathbf{Y} = (Y_1, Y_2, \dots, Y_n)^T$. How do we compute the ML estimate of p ? We will consider three cases separately.

⁶ Recall that $\log(\prod_{i=1}^n \mathcal{L}(\mathbf{t} | Y_i, \mathbf{X}_i)) = \sum_{i=1}^n \log \mathcal{L}(\mathbf{t} | Y_i, \mathbf{X}_i)$.

⁷ It is for this reason that both estimators are special cases of *M-estimators*; see, e.g., Stefanski and Boos (2002).

First, suppose that $\mathbf{Y} = (0, 0, \dots, 0)^T$. Then, $\forall i \in \{1, 2, \dots, n\}$, $\mathcal{L}(\hat{p} | Y_i) = g(Y_i | \hat{p}) = 1 - \hat{p}$, so

$$\mathcal{L}(\hat{p} | \mathbf{Y}) = \prod_{i=1}^n (1 - \hat{p}) = (1 - \hat{p})^n,$$

which, among all $\hat{p} \in [0, 1]$, is clearly maximized at $\hat{p} = 0 = \bar{Y}$.

Now, suppose that $\mathbf{Y} = (1, 1, \dots, 1)^T$. Then, $\forall i \in \{1, 2, \dots, n\}$, $\mathcal{L}(\hat{p} | Y_i) = g(Y_i | \hat{p}) = \hat{p}$, so

$$\mathcal{L}(\hat{p} | \mathbf{Y}) = \prod_{i=1}^n \hat{p} = \hat{p}^n,$$

which, among all $\hat{p} \in [0, 1]$, is clearly maximized at $\hat{p} = 1 = \bar{Y}$.

Finally, suppose that $\exists i \in \{1, 2, \dots, n\}$ such that $Y_i = 0$ and $\exists i' \in \{1, 2, \dots, n\}$ such that $Y_{i'} = 1$. Let $I_0 = \{i : Y_i = 0\}$ and $I_1 = \{i : Y_i = 1\}$. (By assumption, $I_0 \neq \emptyset$ and $I_1 \neq \emptyset$.) Then

$$\mathcal{L}(\hat{p} | \mathbf{Y}) = \prod_{i=1}^n \mathcal{L}(\hat{p} | Y_i) = \prod_{i \in I_0} (1 - \hat{p}) \prod_{i \in I_1} \hat{p} = (1 - \hat{p})^{N_0} \hat{p}^{N_1},$$

where $N_0 = |I_0|$ (the observed number of heads) and $N_1 = |I_1|$ (the observed number of tails).⁸

Notice that $\hat{p} = 0$ or $\hat{p} = 1$ yields $\mathcal{L}(\hat{p} | \mathbf{Y}) = 0$, whereas any $\hat{p} \in (0, 1)$ yields $\mathcal{L}(\hat{p} | \mathbf{Y}) = (1 - \hat{p})^{N_0} \hat{p}^{N_1} > 0$, so $\mathcal{L}(\hat{p} | \mathbf{Y})$ is not maximized at $\hat{p} = 0$ or $\hat{p} = 1$. So, since we know the solution is some $\hat{p}$ such that $\mathcal{L}(\hat{p} | \mathbf{Y}) > 0$, we can take the log-likelihood:

$$\log \mathcal{L}(\hat{p} | \mathbf{Y}) = \log((1 - \hat{p})^{N_0} \hat{p}^{N_1}) = N_0 \log(1 - \hat{p}) + N_1 \log(\hat{p}).$$

The first-order condition is

$$\frac{\partial \log \mathcal{L}(\hat{p} | \mathbf{Y})}{\partial \hat{p}} = 0.$$

This yields

$$-\frac{N_0}{1 - \hat{p}} + \frac{N_1}{\hat{p}} = 0.$$

⁸ We use uppercase letters here since N_0 and N_1 are random variables. Note that, by assumption, $N_0 > 0$ and $N_1 > 0$.

Solving for $\hat{p}$,⁹

$$\begin{aligned}\frac{N_1}{\hat{p}} &= \frac{N_0}{1-\hat{p}} \\ \frac{1-\hat{p}}{\hat{p}} &= \frac{N_0}{N_1} \\ \frac{1}{\hat{p}} - 1 &= \frac{N_0}{N_1} \\ \frac{1}{\hat{p}} &= \frac{N_0}{N_1} + 1 \\ \frac{1}{\hat{p}} &= \frac{N_0 + N_1}{N_1} \\ \frac{1}{\hat{p}} &= \frac{n}{N_1} \\ \hat{p} &= \frac{N_1}{n} \\ \hat{p} &= \frac{1}{n} \sum_{i=1}^n Y_i \\ \hat{p} &= \bar{Y}.\end{aligned}$$

Thus, the ML estimate of the probability of the coin coming up tails is the proportion of flips that came up tails—in other words, the sample mean. That is just the usual plug-in estimate, which corresponds to the sample mean (and, therefore, the regression estimate with no predictors). We know from the Weak Law of Large Numbers (WLLN) that the sample mean is a consistent estimator for p .

To illustrate this more concretely, suppose that we tossed the coin $n = 3$ times and observed $\mathbf{Y} = (1, 0, 1)$. What would be the ML estimate for p ? Figure 5.2.1 shows the likelihood as a function of p given this data. The ML estimate for p corresponds to the maximum in the figure: $\hat{p}_{ML} = \frac{2}{3}$, which is again the plug-in estimate, the proportion of flips that came up tails. $\triangle$

We now turn to the ML estimate of the classical linear model, where a rather remarkable equivalence will be established. This will be a setting where taking the log-likelihood eases the analytic calculations greatly.

⁹ The second-order condition is

$$\begin{aligned}\frac{\partial^2 \log \mathcal{L}(\hat{p} | \mathbf{Y})}{\partial \hat{p}^2} &= \frac{\partial}{\partial \hat{p}} \left(-\frac{N_0}{1-\hat{p}} + \frac{N_1}{\hat{p}} \right) = -(-1) \frac{N_0}{(1-\hat{p})^2} (-1) + (-1) \frac{N_1}{\hat{p}^2} \\ &= -\left(\frac{N_0}{(1-\hat{p})^2} + \frac{N_1}{\hat{p}^2} \right) < 0,\end{aligned}$$

and we already checked the edge cases, so a unique solution will be an absolute maximum.

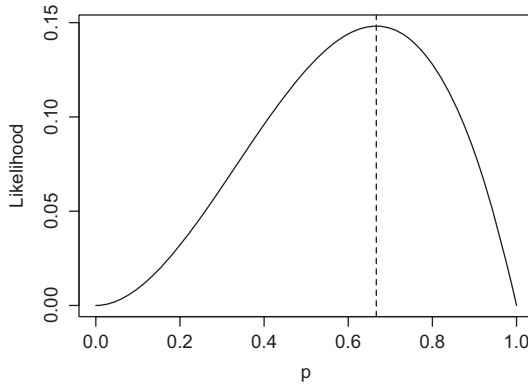


FIGURE 5.2.1. A Likelihood Function for Three Coin Tosses

Example 5.2.4. *Maximum Likelihood Estimation of the Classical Linear Model*

Let Y be a random variable and $\mathbf{X} = (1, X_{[1]}, X_{[2]}, \dots, X_{[K]})$ be a random vector. Suppose that

$$Y = \mathbf{X}\boldsymbol{\beta} + \varepsilon,$$

where $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_K)^T$ and $\varepsilon \sim N(0, \sigma^2)$ conditional on $\mathbf{X}$. Let $(Y_1, \mathbf{X}_1)$, $(Y_2, \mathbf{X}_2)$, ..., $(Y_n, \mathbf{X}_n)$ be i.i.d. observations of $(Y, \mathbf{X})$, and let

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \quad \mathbb{X} = \begin{pmatrix} 1 & X_{[1]1} & X_{[2]1} & \cdots & X_{[K]1} \\ 1 & X_{[1]2} & X_{[2]2} & \cdots & X_{[K]2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{[1]n} & X_{[2]n} & \cdots & X_{[K]n} \end{pmatrix}, \quad \text{and } \mathbf{b} = \begin{pmatrix} b_0 \\ b_1 \\ b_2 \\ \vdots \\ b_K \end{pmatrix}.$$

Then

$$\begin{aligned} \mathcal{L}((\mathbf{b}, s) \mid \mathbf{Y}, \mathbb{X}) &= \prod_{i=1}^n \phi(Y_i, (\mathbf{X}_i \mathbf{b}, s^2)) \\ &= \prod_{i=1}^n \frac{1}{s\sqrt{2\pi}} e^{-\frac{(Y_i - \mathbf{X}_i \mathbf{b})^2}{2s^2}}. \end{aligned}$$

So the log-likelihood is

$$\log \mathcal{L}((\mathbf{b}, s) \mid \mathbf{Y}, \mathbb{X}) = \log \left(\prod_{i=1}^n \frac{1}{s\sqrt{2\pi}} e^{-\frac{(Y_i - \mathbf{X}_i \mathbf{b})^2}{2s^2}} \right)$$

$$\begin{aligned}
 &= \sum_{i=1}^n \log \left(\frac{1}{s\sqrt{2\pi}} e^{-\frac{(Y_i - \mathbf{X}_i \mathbf{b})^2}{2s^2}} \right) \\
 &= \sum_{i=1}^n \left(\log \left(\frac{1}{s\sqrt{2\pi}} \right) + \frac{-(Y_i - \mathbf{X}_i \mathbf{b})^2}{2s^2} \right).
 \end{aligned}$$

We want to find the $\mathbf{b}$ that maximizes $\log \mathcal{L}((\mathbf{b}, s) | \mathbf{Y}, \mathbb{X})$:¹⁰

$$\begin{aligned}
 \hat{\boldsymbol{\beta}}_{ML} &= \arg \max_{\mathbf{b} \in \mathbb{R}^K} (\log \mathcal{L}(\mathbf{b} | \mathbf{Y}, \mathbb{X})) \\
 &= \arg \max_{\mathbf{b} \in \mathbb{R}^K} \sum_{i=1}^n \left(\log \left(\frac{1}{s\sqrt{2\pi}} \right) + \frac{-(Y_i - \mathbf{X}_i \mathbf{b})^2}{2s^2} \right) \\
 &= \arg \max_{\mathbf{b} \in \mathbb{R}^K} \sum_{i=1}^n (-(Y_i - \mathbf{X}_i \mathbf{b})^2) \\
 &= \arg \min_{\mathbf{b} \in \mathbb{R}^K} \sum_{i=1}^n (Y_i - \mathbf{X}_i \mathbf{b})^2.
 \end{aligned}$$

So the ML estimator for $\boldsymbol{\beta}$ is the $\hat{\boldsymbol{\beta}}$ that minimizes the sum of squared residuals. This establishes the formal tie between ML estimation and regression. The ML estimator for $\boldsymbol{\beta}$ is the regression estimator when it is assumed that:

- the linear model is true, and
- the errors are $\varepsilon \sim N(0, \sigma^2)$ conditional on $\mathbf{X}$.

Many textbooks justify OLS under these strong distributional assumptions, but these assumptions are not necessary. Contrast this with our nonparametric results, which require no such distributional assumptions. Under i.i.d. sampling, OLS consistently estimates the best linear predictor (BLP) of the outcome—and of the conditional expectation function (CEF). Δ

It is an unusual feature of these simple cases that there exist closed-form solutions for our ML estimates. (The cases in which they coincide with conventional methods are, naturally, the most amenable to closed-form solutions.) However, in practice, ML estimators rarely have closed-form solutions. Instead, one usually has to calculate the solution numerically using a computer.¹¹ This is the case, for example, for logit and probit. We conclude

¹⁰ We do not need to estimate σ here, but simple algebraic manipulations show that its ML estimator will equal $\frac{1}{n} \sum_{i=1}^n (Y_i - \mathbf{X}_i \hat{\boldsymbol{\beta}})^2$, or the plug-in sample variance of the residuals.)

¹¹ There are many algorithms that can be implemented to find the minimum of a function, generally involving some type of iterated search. For example, the *Newton-Raphson method*

this section by discussing maximum likelihood estimation of the probit model.

Example 5.2.5. Maximum Likelihood Estimation of the Probit Model

Let Y be a binary random variable and $\mathbf{X} = (1, X_{[1]}, X_{[2]}, \dots, X_{[K]})$ be a random vector. Suppose that

$$\Pr[Y = 1 | \mathbf{X}] = \Phi(\mathbf{X}\boldsymbol{\beta}),$$

where $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_K)^T$. This implies that

$$f_{Y|\mathbf{X}}(Y|\mathbf{X}) = g(Y, \mathbf{X}; \boldsymbol{\beta}) = \Phi(\mathbf{X}\boldsymbol{\beta})^Y (1 - \Phi(\mathbf{X}\boldsymbol{\beta}))^{1-Y}.$$

Let $(Y_1, \mathbf{X}_1), (Y_2, \mathbf{X}_2), \dots, (Y_n, \mathbf{X}_n)$ be i.i.d. observations of $(Y, \mathbf{X})$, and let

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \quad \mathbb{X} = \begin{pmatrix} 1 & X_{[1]1} & X_{[2]1} & \cdots & X_{[K]1} \\ 1 & X_{[1]2} & X_{[2]2} & \cdots & X_{[K]2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{[1]n} & X_{[2]n} & \cdots & X_{[K]n} \end{pmatrix}, \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} b_0 \\ b_1 \\ b_2 \\ \vdots \\ b_K \end{pmatrix}.$$

Then

$$\mathcal{L}(\mathbf{b} | \mathbf{Y}, \mathbb{X}) = \prod_{i=1}^n \Phi(\mathbf{X}_i \mathbf{b})^{Y_i} (1 - \Phi(\mathbf{X}_i \mathbf{b}))^{1-Y_i}.$$

Thus, the log-likelihood is

$$\begin{aligned} \log \mathcal{L}(\mathbf{b} | \mathbf{Y}, \mathbb{X}) &= \log \left(\prod_{i=1}^n \Phi(\mathbf{X}_i \mathbf{b})^{Y_i} (1 - \Phi(\mathbf{X}_i \mathbf{b}))^{1-Y_i} \right) \\ &= \sum_{i=1}^n \log \left(\Phi(\mathbf{X}_i \mathbf{b})^{Y_i} (1 - \Phi(\mathbf{X}_i \mathbf{b}))^{1-Y_i} \right) \\ &= \sum_{i=1}^n \left(Y_i \log \Phi(\mathbf{X}_i \mathbf{b}) + (1 - Y_i) \log (1 - \Phi(\mathbf{X}_i \mathbf{b})) \right). \end{aligned}$$

works by taking the second-order polynomial approximation (that is, the Taylor expansion) of the function at a given point by taking numerical derivatives. The first point must be specified as a starting (or “seed”) value. Then it updates to guess the point that maximizes the polynomial approximation. Then it repeats the procedure until it finds a fixed point—a point where the estimate of the maximum does not change by more than a specified tolerance when the procedure is repeated again. Note: depending on the seed value, this method, and many others like it, may not necessarily converge to the global maximum if the function is not strictly concave.

So, given the outcomes for $\mathbf{Y}$ and $\mathbb{X}$, we would obtain the ML estimate by using a computer to calculate

$$\hat{\boldsymbol{\beta}}_{ML} = \arg \max_{\mathbf{b} \in \mathbb{R}^{K+1}} \sum_{i=1}^n \left(Y_i \log \Phi(\mathbf{X}_i \mathbf{b}) + (1 - Y_i) \log(1 - \Phi(\mathbf{X}_i \mathbf{b})) \right). \triangle$$

5.2.2 Maximum Likelihood Estimation when the Parametric Model Is True

Although we do not necessarily believe that the parametric model is true, it will be useful to establish the consistency of the ML estimator when the model is, in fact, true. We provide a slightly informal statement, along with a proof sketch (based on the plug-in principle) for the consistency of the ML estimator.

Theorem 5.2.6. *Consistency of the Maximum Likelihood Estimator*

Let Y be a random variable and $\mathbf{X}$ be a random vector. Let $(Y_1, \mathbf{X}_1)$, $(Y_2, \mathbf{X}_2)$, ..., $(Y_n, \mathbf{X}_n)$ be i.i.d. observations of $(Y, \mathbf{X})$. If the parametric model $f_{Y|\mathbf{X}}(y|\mathbf{x}) = g(y, \mathbf{x}; \boldsymbol{\theta})$ is true and some requisite regularity conditions (such as having a continuous likelihood function, a compact parameter space, and a unique maximum) are satisfied, then $\hat{\boldsymbol{\theta}}_{ML} \xrightarrow{P} \boldsymbol{\theta}$.

Proof sketch: For a more complete derivation, see Wasserman (2004). Let $(Y_1, \mathbf{X}_1)$, $(Y_2, \mathbf{X}_2)$, ..., $(Y_n, \mathbf{X}_n)$ be i.i.d. observations of $(Y, \mathbf{X})$, and let

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} \quad \text{and} \quad \mathbb{X} = \begin{pmatrix} X_{[1]1} & X_{[2]1} & \cdots & X_{[K]1} \\ X_{[1]2} & X_{[2]2} & \cdots & X_{[K]2} \\ \vdots & \vdots & \ddots & \vdots \\ X_{[1]n} & X_{[2]n} & \cdots & X_{[K]n} \end{pmatrix}.$$

Let

$$L_n(\mathbf{t}) = \frac{1}{n} \log \mathcal{L}(\mathbf{t} | \mathbf{Y}, \mathbb{X}).$$

By definition (and the assumption of a unique maximum), $\mathbf{t} = \hat{\boldsymbol{\theta}}_{ML}$ uniquely maximizes $\log \mathcal{L}(\mathbf{t} | \mathbf{Y}, \mathbb{X})$ and therefore uniquely maximizes $L_n(\mathbf{t})$. Now, note that

$$\frac{1}{n} \log \mathcal{L}(\mathbf{t} | \mathbf{Y}, \mathbb{X}) = \frac{1}{n} \log \left(\prod_{i=1}^n g(Y_i, \mathbf{X}_i; \mathbf{t}) \right) = \frac{1}{n} \sum_{i=1}^n \log g(Y_i, \mathbf{X}_i; \mathbf{t}).$$

This is a sample mean, so by the WLLN,

$$\frac{1}{n} \sum_{i=1}^n \log g(Y_i, \mathbf{X}_i; \mathbf{t}) \xrightarrow{P} E[\log g(Y, \mathbf{X}; \mathbf{t})].$$

Let $L(\mathbf{t}) = E[\log g(Y, \mathbf{X}; \mathbf{t})] = E[\log \mathcal{L}(\mathbf{t} | Y, \mathbf{X})]$, so that we can write the above statement as

$$L_n(\mathbf{t}) \xrightarrow{p} L(\mathbf{t}),$$

that is, the sample mean of the likelihood of each observation given parameter values $\mathbf{t}$ converges to the expected likelihood of a single observation given parameter values $\mathbf{t}$.

We now require the following lemma: $\forall \mathbf{t} \in \Theta$,

$$L(\mathbf{t}) \leq L(\boldsymbol{\theta}),$$

with

$$L(\mathbf{t}) = L(\boldsymbol{\theta}) \iff \Pr[g(Y, \mathbf{X}; \mathbf{t}) = g(Y, \mathbf{X}; \boldsymbol{\theta})] = 1.$$

That is, the expected value of the log-likelihood function is uniquely maximized when $\mathbf{t} = \boldsymbol{\theta}$. We omit the proof of this lemma, but essentially, for any $\mathbf{t} \neq \boldsymbol{\theta}$, we know that the distribution associated with $\boldsymbol{\theta}$ would have to fit the true distribution exactly, and any other distribution does not.

So, to complete our sketch, we have shown that

- $\hat{\boldsymbol{\theta}}_{ML}$ uniquely maximizes $L_n(\mathbf{t})$,
- $L_n(\mathbf{t}) \xrightarrow{p} L(\mathbf{t})$, and
- $\boldsymbol{\theta}$ uniquely maximizes $L(\mathbf{t})$.

Given suitable regularity conditions, the ML estimate $\hat{\boldsymbol{\theta}}_{ML} \xrightarrow{p} \boldsymbol{\theta}$, so $\hat{\boldsymbol{\theta}}_{ML}$ is consistent. $\square$

What this proof sketch illustrates is that the ML estimator is itself a type of plug-in estimator. The inferential target, $\boldsymbol{\theta}$, is a feature of the population distribution: the parameter values that maximize the expected value of the log-likelihood function. To estimate this population quantity, we use the sample analog: the parameter values that maximize the sample mean of the likelihood function. Since the sample will more and more closely approximate the population distribution as n grows large, the ML estimate will get closer and closer to the true parameter values.

5.2.3 Maximum Likelihood Estimation when the Parametric Model Is Not True

What happens if the parametric model is wrong (that is, not true)? In such settings, we call the model *misspecified*. What, then, does ML estimation give us, asymptotically under misspecification? The somewhat technical answer is that ML estimation will consistently estimate the parameters that minimize

the *Kullback–Leibler (KL) divergence* of the hypothesized distribution from the true distribution.¹²

Definition 5.2.7. *Kullback–Leibler Divergence for Continuous Distributions*

For probability density functions $p(\cdot)$ and $q(\cdot)$, the *Kullback–Leibler divergence* is

$$\text{KL}[p(\cdot) \| q(\cdot)] = \int_{-\infty}^{\infty} p(t) \log\left(\frac{p(t)}{q(t)}\right) dt.$$

The KL divergence is defined analogously for discrete random variables, with a sum replacing the integral and PMFs replacing PDFs. The Kullback–Leibler divergence is an important idea in multiple fields, including information theory, as it characterizes how well one distribution approximates another distribution.¹³ We can intuitively see this by looking at an extreme case: the KL divergence between any distributions $p(\cdot)$ and $q(\cdot)$ is zero whenever P and Q have identical distributions, as $\frac{p(t)}{q(t)} = 1$ at every point t . For any distributions $p(\cdot)$ and $q(\cdot)$ that differ, the KL divergence will be nonzero and positive.

Theorem 5.2.8. *The ML Estimator Is Consistent for the Minimum KL Divergence Predictor*

Let Y be a random variable and $\mathbf{X}$ be a random vector. Let $(Y_1, \mathbf{X}_1)$, $(Y_2, \mathbf{X}_2)$, ..., $(Y_n, \mathbf{X}_n)$ be i.i.d. observations of $(Y, \mathbf{X})$. Then the ML estimate of $f_{Y|\mathbf{X}}(y|\mathbf{x})$, $g(y, \mathbf{x}; \hat{\theta}_{ML})$ consistently estimates the minimum Kullback–Leibler divergence approximation of $f_{Y|\mathbf{X}}(y|\mathbf{x})$ among all probability distributions of the form $g(y, \mathbf{x}; \mathbf{t})$. That is, assuming that there exists a unique

$$\tilde{\theta} = \underset{\mathbf{t} \in \Theta}{\operatorname{argmin}} \text{KL}[f_{Y|\mathbf{X}}(\cdot | \mathbf{x}) \| g(\cdot, \mathbf{x}; \mathbf{t})],$$

then

$$\hat{\theta}_{ML} \xrightarrow{p} \tilde{\theta}.$$

We omit the proof of this theorem. What is important here is to note that this fact resembles a result we derived for OLS, only with a different metric for the “distance” between distributions (KL divergence rather than MSE).

¹² The parameters that minimize KL divergence are sometimes referred to as the *quasi-ML* parameters.

¹³ The attentive reader may notice that the KL divergence is not symmetric; there exist probability density functions $p(\cdot)$ and $q(\cdot)$ such that $\text{KL}[p(\cdot) \| q(\cdot)] \neq \text{KL}[q(\cdot) \| p(\cdot)]$.

Among all linear predictors, OLS consistently estimates the one “closest” (in terms of MSE) to the CEF. Likewise, among all conditional distributions of the form given by the hypothesized model, ML estimation consistently estimates the one “closest” (in terms of KL divergence) to the true conditional distribution. In short, there exists a sensible definition of “best” under which ML estimation consistently estimates the best-fitting distribution among all distributions in the assumed parametric family.

The idea behind the proof of Theorem 5.2.8 is again related to the plug-in principle. There is a population quantity of interest, $\tilde{\theta}$: the parameter values that minimize the KL divergence of the hypothesized distribution from the true population distribution. It can be shown that the ML estimate is equal to the sample analog: the parameter values that minimize the KL divergence of the hypothesized distribution from the observed data. So, since the distribution of the observed data converges to the population distribution as n grows large, the ML estimate converges to the value that minimizes the KL divergence of the hypothesized distribution from the true population distribution.

If in fact $f_{Y|X}(y|\mathbf{x}) = g(y, \mathbf{x}; \theta)$, then $\text{KL}[f_{Y|X}(\cdot|\mathbf{x}) \| g(\cdot, \mathbf{x}; \theta)] = 0$. Thus, if the model is true, the minimum KL divergence approximation is the true distribution, so $\tilde{\theta} = \theta$, and thus ML estimation consistently estimates the true parameter values, as before. When the parametric model is true, ML asymptotically gets the distribution exactly right. When the parametric model is not true, ML asymptotically gets it “as right as possible.” The following example considers a visualization.

Example 5.2.9. Maximum Likelihood Estimation of an Unusual Distribution

Suppose that the true distribution of Y were given by the histogram on the lefthand side of Figure 5.2.2. Suppose that we erroneously assumed that Y was normally distributed:

$$g(y; (\mu, \sigma)) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y-\mu)^2}{2\sigma^2}}.$$

Then the asymptotic ML estimate of the distribution would be as shown in red on the righthand side of the figure. $\triangle$

A simpler example is the coin flip from Example 5.1.2 (*A Biased Coin Flip*). When we observe $\mathbf{y} = (1, 0, 1)$, we can match that empirical distribution exactly by setting $p = \frac{2}{3}$. With i.i.d. observations, the empirical distribution will converge to the true distribution as n grows large. Thus, asymptotically, the ML estimate will exactly characterize the probability distribution of Y .

Cases like the coin flip, where the model $f_{Y|X}(y|\mathbf{x})$ can be virtually ensured to be true, may be rare in practice, however. In general, ML estimation is best thought of as an explicit means for approximating $f_{Y|X}(y|\mathbf{x})$.

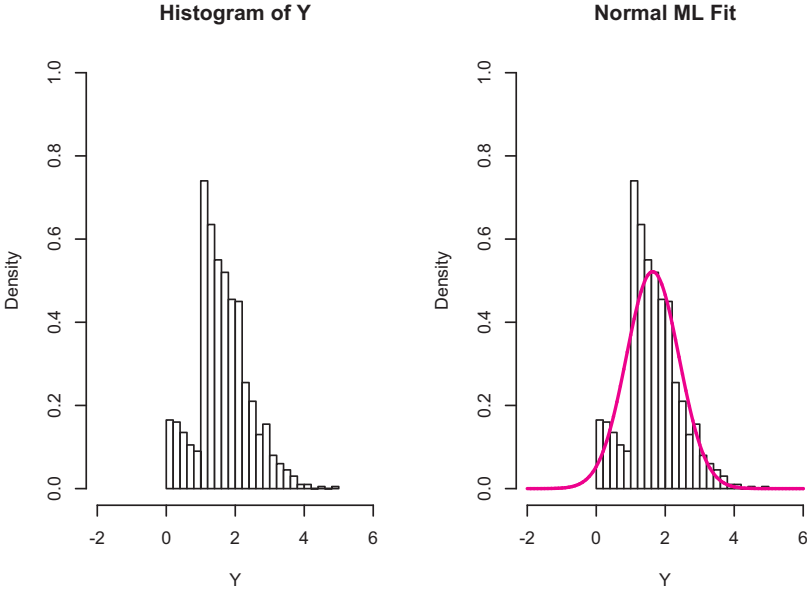


FIGURE 5.2.2. Minimum KL Divergence Approximation of an Unusual Distribution

5.2.4 Maximum Likelihood Plug-In Estimation

Once we conceive of maximum likelihood as an approximation tool, we begin to see its real benefits. Suppose that what we are interested in is not the parameters of the model per se, but instead some other feature of the distribution of $(Y, \mathbf{X})$. We know from Chapter 3 that the joint CDF of Y and $\mathbf{X}$ can be estimated to arbitrary precision. But we can use ML estimation to potentially improve the precision of our estimates by imposing a parametric model that seems likely to approximate the shape of the true CEF reasonably well. As before (Definitions 3.3.3 or 3.3.9), we can formally define a plug-in estimator.

Definition 5.2.10. *Maximum Likelihood Plug-In Estimator*
For i.i.d. random vectors $(Y_1, \mathbf{X}_1), (Y_2, \mathbf{X}_2), \dots, (Y_n, \mathbf{X}_n)$ with common CDF F , the ML *plug-in estimator* of $\psi = T(F)$ is

$$\hat{\psi}_{ML} = T(\hat{F}_{ML}),$$

where

$$\hat{F}_{ML}(y, \mathbf{x}) = \frac{1}{n} \sum_{i=1}^n \left(\mathbb{I}(X_{[1]i} \leq x_{[1]i}, X_{[2]i} \leq x_{[2]i}, \dots, X_{[K]i} \leq x_{[K]i}) \sum_{v \leq y} g(v, \mathbf{X}_i; \hat{\theta}_{ML}) \right)$$

if $f_{Y|X}(y|\mathbf{x})$ is a PMF, and

$$\hat{F}_{ML}(y, \mathbf{x}) = \frac{1}{n} \sum_{i=1}^n \left(I(X_{[1]i} \leq x_{[1]i}, X_{[2]i} \leq x_{[2]i}, \dots, X_{[K]i} \leq x_{[K]i}) \int_{-\infty}^y g(v, \mathbf{X}_i; \hat{\theta}_{ML}) dv \right)$$

if $f_{Y|X}(y|\mathbf{x})$ is a PDF.

The definition follows from invoking the Law of Iterated Expectations and substituting sample quantities for population analogs. To see the ML plug-in estimator in action, let us consider the CEF of Y given $\mathbf{X} = \mathbf{x}$. Given the definition of a conditional expectation (Definition 2.2.10),

$$E[Y|\mathbf{X} = \mathbf{x}] = \int_{-\infty}^{\infty} yg(y, \mathbf{x}; \theta) dy.$$

The ML plug-in estimator is then obtained by substituting the ML estimate, $\hat{\theta}_{ML}$, for the true values of θ :

$$\hat{E}[Y|\mathbf{X} = \mathbf{x}] = \int_{-\infty}^{\infty} yg(y, \mathbf{x}; \hat{\theta}_{ML}) dy.$$

As usual, in practice, we do not always have to go this far. For example, in the case of the CEF of the classical linear model (Theorem 5.1.5), we need only use $\hat{E}[Y|\mathbf{X} = \mathbf{x}] = \mathbf{x}\hat{\beta}_{ML}$; and in the case of the CEF of a binary choice model (Theorem 5.1.8), we need only use $\hat{E}[Y|\mathbf{X} = \mathbf{x}] = h(\mathbf{x}\hat{\beta}_{ML})$.

If the parametric model is true, then under the usual plug-in regularity conditions, $\hat{E}[Y|\mathbf{X} = \mathbf{x}]$ will be consistent for the CEF evaluated at $\mathbf{X} = \mathbf{x}$. If the parametric model is not true, then our estimator of the CEF will have no guarantees of consistency. However, insofar as the parametric model is a good approximation, the ML plug-in estimator can be expected to behave well.

5.2.5 Mixture Models

We can use an extension of ML estimation, where we will allow for the possibility that *multiple* distributions give rise to the data. This approach is known as *mixture modeling*. While one model may fail in approximating Y well, many may succeed.

Suppose that the distribution of Y conditional on $\mathbf{X}$ can be represented as a weighted average of M different parametric models,

$$\begin{aligned} f_{Y|X}(y|\mathbf{x}) &= g(y, \mathbf{x}; \theta) \\ &= p_1 g_1(y, \mathbf{x}; \theta_1) + p_2 g_2(y, \mathbf{x}; \theta_2) + \dots + \left(1 - \sum_{m=1}^{M-1} p_m \right) g_M(y, \mathbf{x}; \theta_M), \end{aligned}$$

where $\theta = (p_1, \theta_1, p_2, \theta_2, \dots, p_{M-1}, \theta_{M-1}, \theta_M)$. Then, as before, $\prod_{i=1}^n g(Y_i, \mathbf{X}_i; \theta)$ can be maximized to find the ML estimate of θ . (This can sometimes be computationally tricky.)

If any subset of the models in $(g_1(y, \mathbf{x}; \theta_1), \dots, g_M(y, \mathbf{x}; \theta_M))$ jointly characterize the conditional distribution of Y , then this estimator is consistent and asymptotically efficient. When the models are not valid (even jointly), our estimates should be interpreted explicitly as an approximation.

Example 5.2.11. Mixture Model Approximation

Consider again the unusual distribution discussed in Example 5.2.9 (*Maximum Likelihood Estimation of an Unusual Distribution*), visualized in Figure 5.2.3.

In Example 5.2.9, we showed the consequences of modeling this distribution as normal:

$$g_N(y; (\mu_N, \sigma_N)) = \frac{1}{\sigma_N \sqrt{2\pi}} e^{-\frac{(y-\mu_N)^2}{2\sigma_N^2}}.$$

This model was not true, but nevertheless a useful first-order approximation. But let us consider some other models to approximate this distribution. For

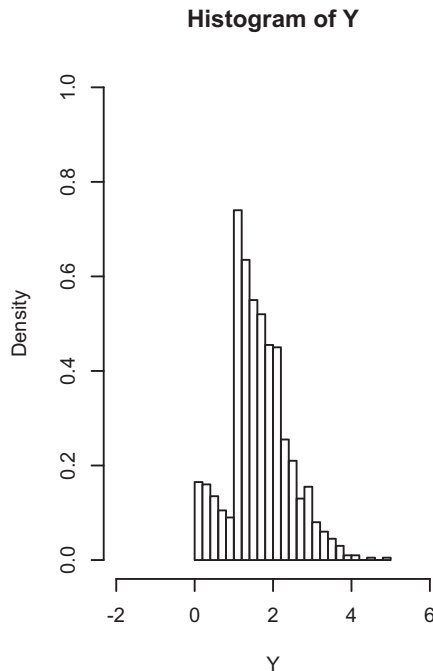


FIGURE 5.2.3. *Unusual Distribution*

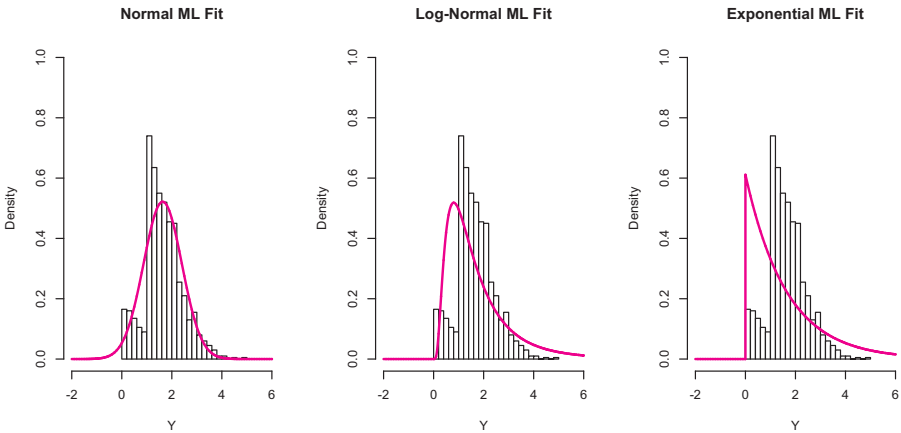


FIGURE 5.2.4. Normal, Log-Normal, and Exponential Maximum Likelihood Approximations

example, we could model the distribution as *log-normal*:

$$g_{LN}(y; (\mu_{LN}, \sigma_{LN})) = \frac{1}{y\sigma_{LN}\sqrt{2\pi}} e^{-\frac{(\log y - \mu_{LN})^2}{2\sigma_{LN}^2}},$$

or we could model the distribution as *exponential*:

$$g_E(y; \lambda_E) = \lambda_E e^{-\lambda_E y}.$$

The ML estimates from each of these models are visualized in Figure 5.2.4.

As we can see, all these estimates perform relatively poorly in characterizing the true distribution. But what happens when we combine them, and allow for the distribution of Y to be an optimal combination of the three distributions?

Consider a mixture model that incorporates all three models:

$$\begin{aligned} g(y; \theta) &= p_N g_N(y; (\mu_N, \sigma_N)) + p_{LN} g_{LN}(y; (\mu_{LN}, \sigma_{LN})) \\ &\quad + (1 - p_N - p_{LN}) g_E(y; \lambda_E) \\ &= p_N \frac{1}{\sigma_N \sqrt{2\pi}} e^{-\frac{(y - \mu_N)^2}{2\sigma_N^2}} + p_{LN} \frac{1}{y\sigma_{LN}\sqrt{2\pi}} e^{-\frac{(\log y - \mu_{LN})^2}{2\sigma_{LN}^2}} \\ &\quad + (1 - p_N - p_{LN}) \lambda_E e^{-\lambda_E y}, \end{aligned}$$

where $\theta = (p_N, p_{LN}, \mu_N, \sigma_N, \mu_{LN}, \sigma_{LN}, \lambda_E)$. Figure 5.2.5 visualizes the ML estimate of this mixture model, which provides a markedly better approximation. Δ

Technically, mixture models are parametric, as the implied distribution of Y given $\mathbf{X}$ is fully governed by a finite number of parameters. But ML estimates of mixture models often have the operating characteristics of nonparametric

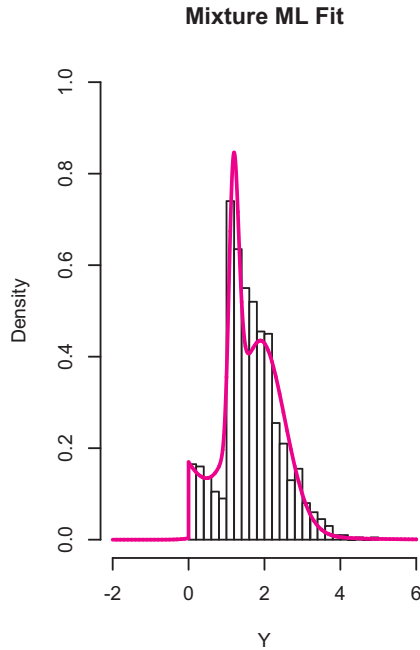


FIGURE 5.2.5. *Mixture Model Maximum Likelihood Approximation*

estimators, and may often outperform explicitly nonparametric methods, including kernel methods. There is, of course, a cost: using a mixture model in this way increases the complexity of the model relative to simpler ML specifications. Accordingly, when using mixture models with small n , we may suffer from overfitting (Section 4.3.2) from trying to be too flexible in approximating the data.

Depending on how we conceptualize their asymptotics, ML estimates of mixture models can be thought of as nonparametric estimators. If we imagined a sieve estimator where we added more and more models in a systematic way as n grew large in a sufficiently general way, then this would result in an infinite-dimensional model that could approximate any arbitrary distribution. Of course, once we have our ML estimate of the density function $f_{Y|X}(y|\mathbf{x})$, we again can do whatever we want with it, including characterizing the CEF, partial derivatives, or any other distributional feature.

5.2.6 Penalized Maximum Likelihood Regression

As with linear regression, it is possible to generalize ML estimation to allow for penalization for model complexity. Analogously to our discussion of penalized regression in Section 4.3.6, this can be formalized by introducing a penalty

term and minimizing the sum of the penalty and the negative log-likelihood. Penalized ML estimation is important in controlling model overfitting, even of the type introduced by mixture models.

Here we discuss one such approach: ℓ^1 -penalized ML estimation, or the analog of the lasso in the domain of ML estimation. Mirroring the language of Section 4.3.6, it is a way of flexibly estimating the conditional density function when we have a very large number of parameters to estimate. The ℓ^1 -penalized ML estimate,

$$\hat{\theta} = \underset{\theta \in \Theta}{\operatorname{argmin}} \left(-\log \mathcal{L}(\theta | \mathbf{Y}, \mathbb{X}) + \lambda \|\theta^P\|_1 \right),$$

where θ^P represents a subvector of θ corresponding to model parameters subject to penalty; $\|\cdot\|_1$ is again the ℓ^1 norm,

$$\|\theta^P\|_1 = \sum_{k=1}^{|\theta^P|} |\theta_k^P|,$$

which is the sum of the absolute value of the model parameters subject to penalty; and λ is again a penalty parameter. Typically, θ^P contains parameters whose scale directly corresponds to model complexity (for example, elements of β in the classical linear model), but not parameters whose scale does not correspond to model complexity (for example, σ^2 in the classical linear model).

We want to keep the model parameters close to zero, unless they really help to maximize the log-likelihood. Subject to the constraints of the parametric model being used, penalization presents a way to use many explanatory variables to “automatically” get good estimates of the conditional outcome distribution—including when we have more variables than observations. The interpretation of penalized ML estimates proceeds exactly as with ML estimates, and plug-in estimates of, for instance, average partial derivatives can be straightforwardly obtained.

As with the case of penalized regression, choosing λ is not easy, and there is no general best way to estimate it. As with the lasso, cross-validation (see Section 4.3.6 for details) often has good operating characteristics. As long as we assume that, as $n \rightarrow \infty$, λ does not grow too quickly, then the ℓ^1 -penalized estimator will usually be consistent for the minimum MKLD Predictor (see Section 5.2.3). Other penalization methods exist—ridge-type (ℓ^2 -penalized) ML estimates and penalized mixture models—and all operate under the same principles.

5.2.7 Inference

A remaining question is how to characterize uncertainty for our ML estimates. There is debate over whether or not robust standard errors and/or

the bootstrap are appropriate for parametric models. It is possible to estimate standard errors that exploit all the parametric assumptions that we have used to generate the model. If we have estimates of the full distribution of Y conditional on X , we can use that information to construct a standard error estimate. If the model is true, then we can directly use the model to estimate the standard errors. This approach, like classical standard errors for regression (Section 4.2.3), is valid if strong assumptions hold. Here, model-based standard errors depend on model validity. That is, if the parametric model is not true, model-based standard errors are not generally consistent.

Fortunately, the bootstrap and robust standard errors (the form of which we do not detail here) are available. In large samples, under the usual plug-in regularity conditions, they accurately characterize the sampling variability of our estimates, even when the model is not true. While parametric models are almost certainly wrong, the estimates produced by ML estimation may still be meaningful as approximations. In Section 4.2.3, we considered “classical sampling variance estimation” for the OLS estimator—an estimator that is logically equivalent to the classical standard error for the ML estimate of the classical linear model. The estimates associated with OLS and other maximum likelihood solutions may have value as approximations even when the parametric model is not true. We should not needlessly run the risk of misstating the sampling variability of these estimates. Given the possibility that any given parametric model is not true, for the types of problems that applied researchers face, robust standard errors or the bootstrap are typically preferable.

As long as we do not take the parametric model too seriously, ML estimation can be construed as a principled way of approximating a probability distribution. When we are interested in the features of the probability distribution, then the ML estimates are best construed as approximations to these features. Robust standard errors and the bootstrap guarantee, in large samples, that we are accurately quantifying the sampling variability of these approximations. Model-based standard errors offer no such guarantee.

5.3 A NOTE ON MODELS AS APPROXIMATIONS

We began this chapter with a famous quote by the statistician George Box. Many texts in statistics and econometrics include this quotation. In most cases, however, it is essentially taken to justify a more or less cavalier approach to statistical models, one that casually treats them as though they were true.

In this chapter, we have adhered to a more conservative interpretation of Box’s aphorism. All models are wrong; they are, at best, approximations of reality. But, even without assuming that they are exactly true, when employed and interpreted correctly, they can nonetheless be useful for obtaining estimates of features of probability distributions.

5.4 FURTHER READINGS

Cameron and Trivedi (2005) contains an excellent treatment of the properties of ML and other estimators when the motivating model is wrong. Similarly, Stefanski and Boos (2002) is a very clear, somewhat technical, article-length exposition of this topic. For more traditional approaches under a structural paradigm, we recommend the following textbooks, all of which are comprehensive, clear and highly readable: Hayashi (2001), Wooldridge (2010), and Greene (2012). For a more skeptical treatment of many of the same materials, we strongly recommend the elegant, clear, and canonical Freedman (2009).